# Homework 8

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**Problem 1.** Write out either a self-reference template proof or a diagonalization template proof that L is not decidable. TM H is now a TM that decides language L.

*Proof.* .
Given:
L* = { < M >, w: M accepts w, and at termination the first character on the tape for M is 0.}.
$\Sigma$ = {0,1}

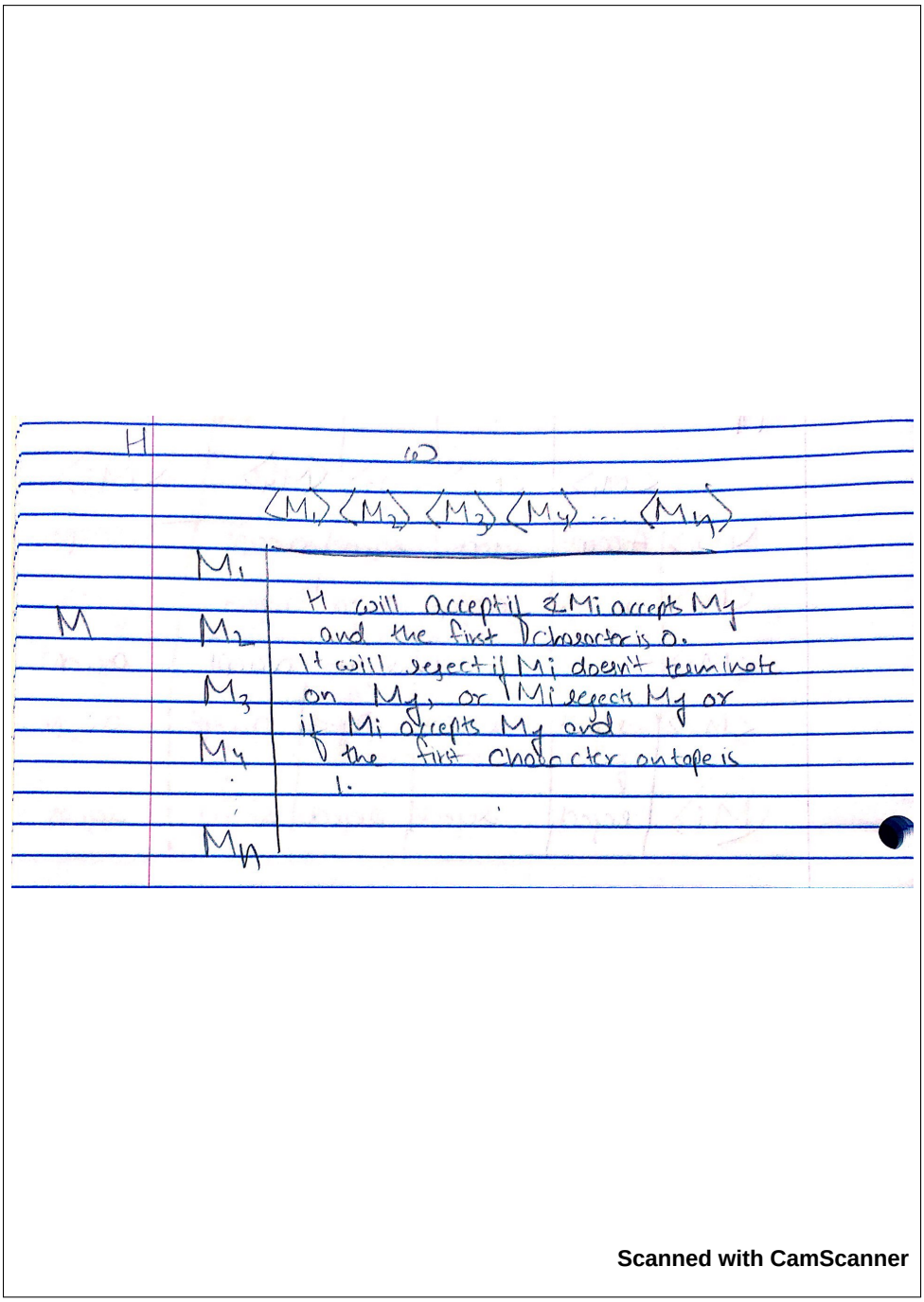
To prove that:
L* is undecidable.

Proof:

Lets assume we have a decider H, for the language L*.
We have multiple cases for L*. Lets look at the end cases and the behavior of H for these cases:
1)M accepts w and the first character on its tape at termination is 0. H Accepts
2)M accepts w and the first character on its tape at termination is 1. H Rejects
3)M rejects w and the first character on its tape at termination is 0. H Rejects
4)M rejects w and the first character on its tape at termination is 1. H Rejects
5)M does not terminate when given input w. H Rejects

Lets give input to H, of the form < $M_i$, < $M_j$ > >. Where M $_i$ and $M_j$ are same or different than each other.
Lets look at the behavior of H for the input < $M_i$, < $M_j$ > > for **ALL TURING MACHINES**.

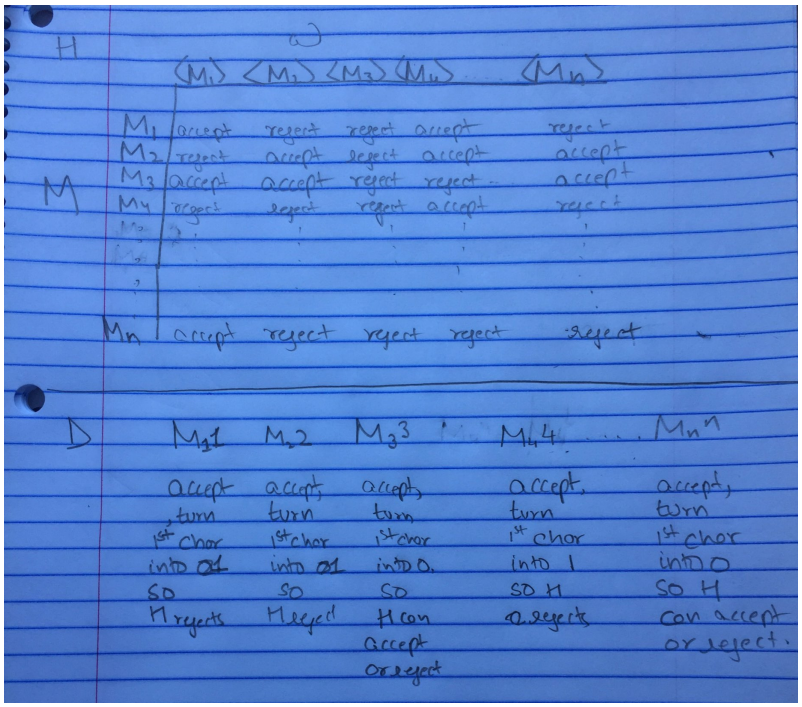
# 1 TABLE A




Lets have a new TM D, which does the following:
If H accepts the string < M >, < M >
{write 1 on the first tape cell and accept}
else:
{write 0 on the first tape cell and accept}.

Lets look at the behavior of D given a table for H.

## 2 TABLE B



H

| M | $\langle M_1 \rangle$ | $\langle M_2 \rangle$ | $\langle M_3 \rangle$ | $\langle M_4 \rangle$ | ... $\langle M_n \rangle$ |
|---|---|---|---|---|---|
| $M_1$ | accept | reject | reject | accept | reject |
| $M_2$ | reject | accept | reject | accept | accept |
| $M_3$ | accept | accept | reject | reject | accept |
| $M_4$ | reject | reject | reject | accept | reject |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| $M_n$ | accept | reject | reject | reject | reject |

D

| $M_1 1$ | $M_2 2$ | $M_3 3$ | $M_4 4$ ... | $M_n n$ |
|---|---|---|---|---|
| accept, turn 1st char into 1 so H rejects | accept, turn 1st char into 1 so H rejects | accept, turn 1st char into 0. so H can accept or reject | accept, turn 1st char into 1 so H rejects | accept, turn 1st char into 0 so H can accept or reject. |

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Since D is a TM, then D has to be somewhere in TABLE A, since it is a table for ALL TURING MACHINES.
Lets assume D corresponds to row k.
Lets look at the behavior $< M_k , < M_k >>$ i.e. running TM D on TM D.
If the table for H has an accept for $< M_k , < M_k >>$ then the way D is constructed it will turn the first character of the input string into a 1 which should be a reject for H since H rejects anything whose first character is not a 0 . This is impossible since H cannot be accept and reject at the same time. Since k is an arbitrary row, then D cannot exist anywhere in the table for H. Therefore, H cannot decide for a TM D, and thus the decider H cannot exist.

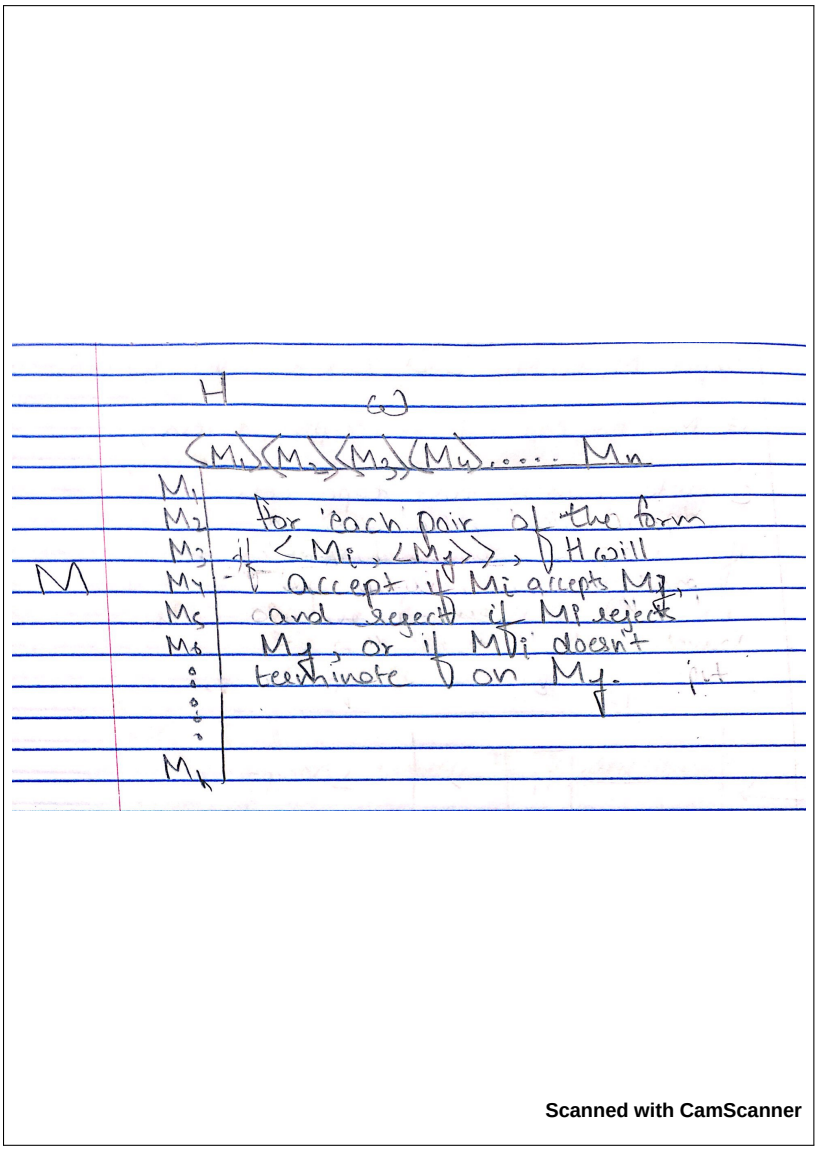
Therefore, our initial assumption was false, and L* is not decidable.

$\square$

**Problem 2.** Write out in your own words a full detailed (what I call a template) proof that the language ATM is undecidable, using a table of behaviors, and diagonalization. This is sketched in the book, but the template proof should be more complete.

*Proof.* .
**To prove that:**
ATM = $\{< M, w > \mid$ M is a TM and M accepts w$\}$ is undecidable.
**Proof**
Lets assume that ATM is decidable.
If ATM is decidable, the we have a decider, H for ATM that can, on given an input $< M, w >$ will halt and accept if w is in M, or reject if w is not in M.(Since it is a decider it will not go on forever)
For w to ATM, we let w be a description of a TM, as binary strings.
We know the above can be done as we learned in class.
Given a TM in the form of a binary string, and an input,we can simulate it on a UTM, so we know this can be done.
In our case, the input is the description of the TM as we have mentioned earlier.
Therefore, lets look at the table for **ALL TURING MACHINES** for H for the input of the form ($M_i$, $M_j$), where $M_i$, $M_j$ are turing machines and it can be the same as each other or different than each other.

# 3 Table 1



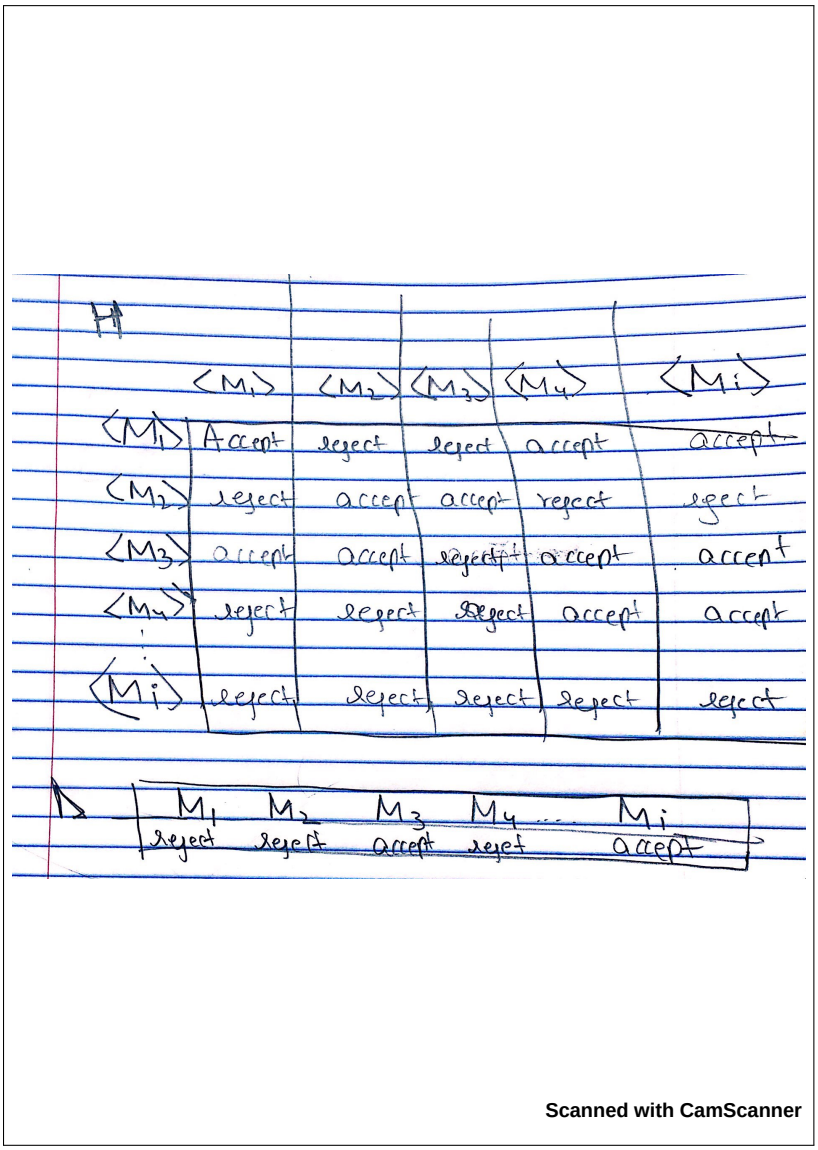

NOTE: It is completely possible that some of the machines are invalid descriptions of TMs, in this case H will reject the input.

Lets have a new TM D, which looks at the result of H on a pair $< \mathrm{M}_i, < M_i > >$ and flips the result of H.

For example:

If we had a table of this form for H, this is what we would have for D.

# 4 Table 2



H

| | $\langle M_1 \rangle$ | $\langle M_2 \rangle$ | $\langle M_3 \rangle$ | $\langle M_4 \rangle$ | $\langle M_i \rangle$ |
|---|---|---|---|---|---|
| $\langle M_1 \rangle$ | Accept | reject | reject | accept | accept |
| $\langle M_2 \rangle$ | reject | accept | accept | reject | reject |
| $\langle M_3 \rangle$ | accept | accept | reject | accept | accept |
| $\langle M_4 \rangle$ | reject | reject | reject | accept | accept |
| $\langle M_i \rangle$ | reject | reject | reject | reject | reject |

D

| $M_1$ | $M_2$ | $M_3$ | $M_4$ | ... | $M_i$ |
|---|---|---|---|---|---|
| reject | reject | accept | reject | | accept |

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Since, D is a TM, at some point in the table 1, there has to be a row in table 1 since table 1 is a table for all TURING MACHINE, that corresponds to the the one dimensional table we have for D.
Lets say row k in table for H corresponds to the row for D, therefore, for $< M_k, M_k >$, if we have reject in table H, then the way D is constructed it should be accept and vice versa.
But this is impossible, since H cannot be accept and reject at the same time.
Therefore, now row k, for any arbitrary k, cannot correspond to a row in the TABLE FOR ALL TURNING MACHINES
If D is a TM and it cannot exist in the table 1 ( Table for H), then H cannot decide for a TM D and therefore, H cannot exist .
Therefore, our initial assumption was false and ATM is undecidable. □